KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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Software engineer with production backend/full-stack experience and hands-on work building and evaluating LLM systems across Text-to-SQL, hybrid retrieval, and tool-using multimodal agents.

EXPERIENCE

Kahap Inc | *Software Developer (Part-time)* Aug 2022 - Aug 2026

- Primary developer for a production feedback platform, owning submission workflows, automated notifications, and administrative features from implementation through production support.
- Maintained and extended Laravel/Vue.js production applications, shipping business features and debugging application and database issues.
- Handled deployments and server operations, including upgrades, Laravel queues/workers, SSL maintenance, and database troubleshooting.

Pac Com Solution | *Full-Stack Developer Intern* Jun 2021 - Jan 2022

- Developed and maintained full-stack features for production POS applications using Laravel and Vue.js.

PROJECTS

Semantic Text-to-SQL Agent | *Python · LangGraph · Qwen3.5 4B · FastAPI · SQLite*

- Built a local Text-to-SQL service with deterministic SQL verification, read-only execution, bounded repair, and one-pass clause-guided semantic review.
- On a frozen 100-query BIRD DEV run, guarded repair recovered 19 failed executions (63% → 82% execution success); official BIRD EX rose from 31% to 34%.

Local RAG Search | *Python · BM25 · Qwen Embeddings · RRF · FastAPI*

- Built a local hybrid retriever with BM25, Qwen dense retrieval, RRF fusion, and per-result rank provenance.
- On 64 Confluence-compatible queries over 5,189 documents, hybrid RRF reached 0.7630 Recall@10, 0.7418 MRR@20, and 0.8438 Hit@10.

Hermes Video Tutor | *Python · Hermes · Qwen3.5 4B · Tool Calling · Multimodal*

- Built a local video-QA agent that uses transcript evidence first and inspects frames/clips only when needed, with visible tool traces and abstention.
- On a frozen 12-question self-authored paired fixture, multimodal tools raised visual-required full pass from 0% to 66.7%, with no unnecessary visual calls on transcript-answerable cases.

SKILLS

AI / LLM: LLM Agents, RAG, Tool Calling, Retrieval & LLM Evaluation, Embeddings, LangGraph
Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js, React
Cloud / Data: AWS, MySQL, Redis, SQLite, Linux
Other: Java, C/C++, Git, pytest | Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | *M.S., Information Systems & Applications* 2024 - 2026
GPA 4.09/4.3 | Taiwan | Thesis: *Interpreting Implicit Intent through Workplace Hierarchies and Context*

Chung Hua University | *B.S., Computer Science & Information Engineering* 2022 - 2024
GPA 3.9/4.0 | Taiwan